

Quiz (Habanero)

- Q1.** The temperature t in a greenhouse is bounded by $t \geq 60$ and $t \leq 80$. What is the feasible region?
- (A) $t = 60$
 (B) $t = 80$
 (C) $60 \leq t \leq 80$
 (D) $t > 60$ and $t < 80$
 (E) $t < 60$ and $t > 80$
- Q2.** A city's average annual rainfall r is between 20 and 30 inches. Which inequality represents this?
- (A) $r > 20$
 (B) $r < 30$
 (C) $20 < r < 30$
 (D) $20 \leq r \leq 30$
 (E) $r \geq 20$ and $r \leq 30$
- Q3.** What does the overlapping region of $x + y \geq 4$ and $x - y \leq 2$ represent?
- (A) The solution set of the system
 (B) The feasible region for temperature and rainfall
 (C) The intersection of two lines
 (D) The union of two inequalities
 (E) The complement of the solution set
- Q4.** For the system $2x + 3y \geq 12$ and $x - 2y \leq -3$, what is the relationship between x and y in the feasible region?
- (A) $x > y$
 (B) $x < y$
 (C) $x = y$
 (D) $2x + 3y \geq 12$ and $x - 2y \leq -3$
 (E) $2x + 3y \leq 12$ and $x - 2y \geq -3$
- Q5.** If $y \leq 2x - 1$ and $y \geq x + 1$, what is the range of y when $x = 2$?
- (A) $3 \leq y \leq 4$
 (B) $y = 3$
 (C) $y \geq 3$
 (D) $3 \leq y \leq 5$
 (E) $y \leq 3$
- Q6.** The feasible region for the system $x + 2y \geq 6$ and $3x - y \leq 7$ is a
- (A) Triangle
 (B) Rectangle
 (C) Polygon
 (D) Circle
 (E) Line segment
- Q7.** What is the x -coordinate of the point of intersection between $y = 2x - 1$ and $y = x + 1$?
- (A) 2
 (B) 1
 (C) 4
 (D) 0
 (E) 3
- Q8.** For the system $y \leq -x + 3$ and $y \geq 2x - 2$, which of the following points is in the feasible region?
- (A) (1, 1)
 (B) (2, 2)
 (C) (0, 0)
 (D) (1, 2)
 (E) (3, 0)

Open Ended Questions (FRQ)

- Q9.** Graph the system $x + y \geq 2$ and $x - y \leq 1$. Identify the vertices of the feasible region and determine the range of y when $x = 1$. Show your work and explain your reasoning.
- Q10.** The population P of a certain species of bird is affected by the amount of rainfall r and the average temperature t . The population is modeled by the system $P \leq 100 + 2r - t$ and $P \geq 50 + r + t$. If the average temperature is $65^\circ F$ and the rainfall is 10 inches, find the range of the population and explain the feasible region. Show your work and provide a brief explanation.

Chef's Tips

- Tip 1: Ensure students have graph paper and a pencil for graphing the inequalities
- Tip 2: Encourage students to check their work by plugging in values from the feasible region into the inequalities
- Tip 3: Remind students to provide clear explanations and show their work for the free response questions